

Ángel Rafael Rubio Fernández

Embedded Software Engineer | C/C++, Linux & VxWorks | Requirements, Verification & Real-Time Defence/Aerospace Software

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PROFESSIONAL PROFILE

Embedded Software Engineer with 7+ years in software engineering, including 5+ years in embedded, firmware, defence and aerospace environments. Hands-on experience with C/C++, C++17, Linux, VxWorks and requirements-driven development across ESA/CNE-related systems, defence radar software, A400M simulation software at Airbus Defence and Space, and industrial firmware at HP. Focused on traceability, DOORS-based requirements workflows, automated verification, real-time/RTOS environments, software configuration/change management and high-reliability engineering delivery.

CORE COMPETENCIES

Embedded and real-time: C/C++, C++17, Embedded C++, Linux/Embedded Linux, VxWorks, Yocto, RTOS, multithreading, real-time simulation, low-level development and target integration.

Lifecycle and verification: software requirements, design, coding, verification, integration, configuration/change management, documentation release support, software QA, DOORS, traceability and regression control.

Testing and tools: automated testing, unit testing, GoogleTest, Rhapsody, Git, CMake, Jira, Jenkins, Python, TCL and Shell scripting.

Standards and practices: secure coding, safety-critical development practices, MISRA awareness, DO-178C awareness and high-reliability engineering environments.

Domains and stakeholders: defence/aerospace systems, radar, airborne real-time software, A400M, ESA/CNE; Product/Program Managers, System Engineers, Validation teams, QA and software stakeholders.

PROFESSIONAL EXPERIENCE

Senior Embedded Software Engineer

Cipherbit-Grupo Oesía | Sep 2025 - Present | Madrid, Spain

- Develop embedded software in C/C++ and Linux-oriented environments for defence and aerospace-related systems, including ESA/CNE-related projects.
- Work across software requirements, design, coding, verification, integration and technical documentation in high-reliability engineering contexts.
- Use DOORS-centred requirements and traceability workflows to keep implementation, verification evidence and documentation aligned.
- Apply secure coding, safety-critical development practices, MISRA/DO-178C-aware engineering and disciplined configuration/change management.
- Collaborate with Product/Program Managers, System Engineers, Validation teams, QA and software stakeholders to support predictable delivery.

Defence Embedded Software Engineer

Bertrandt Technology Spain / Indra Sistemas | Sep 2024 - Sep 2025 | Madrid, Spain

- Developed embedded C++ software for defence radar systems, focusing on requirements traceability, integration, verification and maintainable delivery.
- Designed and implemented an automated testing library for radar requirements verification, reducing manual testing effort and improving repeatability and reuse of verification assets.
- Worked with C++17, Linux, VxWorks, Yocto, DOORS, Rhapsody, Jira, Git, CMake, GoogleTest, Python, TCL and Jenkins across development and verification workflows.

Software Architect / Aerospace & Defence Software Engineer

Capgemini Engineering / Airbus Defence and Space | May 2023 - Sep 2024 | Madrid, Spain

- Designed software architecture and developed C++ software for requirements verification of the HF module of the A400M simulator at Airbus Defence and Space.
- Implemented a multithreaded architecture in a Linux-based real-time simulation environment, supporting robust verification and regression workflows.
- Used DOORS-based requirements traceability to connect software behaviour, verification activities and documentation in an aerospace/defence context.
- Developed unit-testing and functional verification approaches to improve maintainability, regression control and evidence generation.

Firmware Engineer

HP SCDS / HP Inc. | Jan 2021 - Apr 2023 | León, Spain

- Acted as Platform Owner in a Scrum environment for HP Latex 700/800 and DesignJet Z6/Z9 Pro printers within Current Product Support.
- Coordinated firmware activities, official release support, version-control integration, software configuration/change management and stakeholder communication.
- Investigated and corrected firmware defects inherited from New Product Development phases.
- Delivered C++ design-pattern training at HP School 2022 for future firmware engineers.

ADDITIONAL EXPERIENCE

Software Developer / VR, AR and Simulation

University of Extremadura / GexCALL / FunDECYT-PCTEEx | 2018 - 2020 | Extremadura, Spain

- Developed simulation, VR/AR and interactive software prototypes using Unity/C#, computer vision and applied engineering workflows.

Industrial Design Intern

COVELESS Ingeniería | Feb 2018 - Apr 2018 | Badajoz, Spain

- Worked on parametric 3D modelling with Autodesk Inventor for a universal robotic palletizing tool.

EDUCATION

MSc in Simulation in Science and Engineering

University of Extremadura | 2018 - 2020

Thesis: automatic license plate recognition system based on computer vision.

BSc in Electronics and Automation Engineering

University of Extremadura | Industrial Engineering branch | 2014 - 2018

Final project: control system for unmanned aerial vehicles using voice commands. Awarded Best Final Degree Project 2018/2019 by the Official College of Industrial Technical Engineers of Cáceres.

LANGUAGES

Spanish: native. English: C1.

SELECTED PUBLICATIONS & INTERNATIONAL EXPERIENCE

- **2019:** Usability and Face Validity of the Virtual Soft Skills Game - CASEIB, Santander.
- **2018:** Development of a control system for unmanned aerial vehicles using voice commands - XXXIX Jornadas de Automática.
- **International:** Erasmus+ S4Game, Czech Republic; Erasmus+ International Debate Exchange, Bulgaria; Camp Rising Sun, USA.